

ADC-aware PTQ and Post-ADC Low-Rank Correction for Llama

From Dead-Zone Diagnosis to INT4 $\approx$ INT8 PTQ Quality

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Problem: ADC Severely Degrades Low-Bit LLM Inference

The opportunity

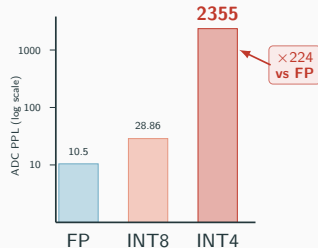
- Analog MVM is 10–100× more energy-efficient than digital
- Attractive for LLM inference at scale

The bottleneck: ADC

- Each tile output must be digitised by an ADC with step δ
- Small outputs fall into the **ADC dead band**
- Under INT4, most outputs are small $\Rightarrow$ **quality collapse**

Goal

Recover LLM quality under a **frozen analog hardware model** — no changes to ADC circuitry or tile structure.



Research Questions

Q1 Diagnosis.

Is the dead-zone collapse hardware-fundamental, or a transform-training artefact?

Q2 INT8 recovery.

How much can ADC-aware PTQ recover compared to the FP baseline?

Q3 INT4 pure PTQ.

What is the best calibration recipe under INT4 ADC constraints?

Q4 Breaking the plateau.

If PTQ saturates, where and how should residual correction be inserted?

Our approach

1. Improve FlatQuant training
2. Propagated ADC-aware PTQ (α -mixed loss, staged diag)
3. Post-ADC residual LoRA with CE+KL teacher

Q1 → Slide 7

Q2 → Slide 8

Q3 → Slides 9–10

Q4 → Slides 12–14

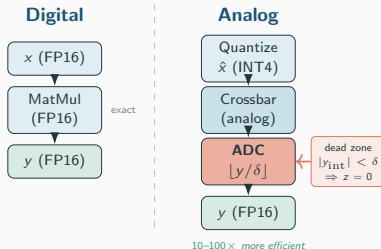
Why Analog Inference? Why This Is Hard.

The opportunity: analog MVM

- LLM inference is memory- and compute-bound
- Analog *multiply-accumulate* is 10–100× more energy-efficient vs. digital
- Real benefit only if output precision is sufficient

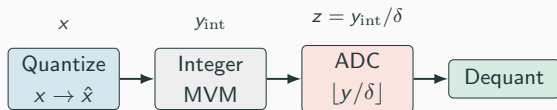
The bottleneck: ADC

- After each tile MVM, an ADC digitises the analog accumulator
- ADC resolution is *coarse*: $\delta \approx 2016$ (INT8 tile)
- Poor activation distributions $\Rightarrow$ most outputs fall in the **dead zone** ($|y_{\text{int}}| < \delta$)



Hardware Model & Metrics

One tile of analog MVM



Dead zone: $|y_{\text{int}}| < \delta \Rightarrow z = 0$ (ADC outputs nothing)

$$\delta = \frac{2 \text{ tile_in} \cdot q_x q_w}{2^{b_a} \cdot k} \quad \text{INT8 } (q=127): \approx 2016; \text{ INT4 } (q=7): \approx 6.12$$

Key metrics

- $\text{PPL}_{\text{bypass}}$ — PPL with ADC disabled (transform quality proxy)
- PPL_{ADC} — PPL with full ADC pipeline (what matters)
- `dead_rate` — fraction of channels with $z = 0$

Parameter	Value
Model	Llama-3.2-1B
b_w, b_x	4 or 8 bits
b_a (ADC)	8 bits
Tile size	256
k (parallelism)	16
δ (INT4)	≈ 6.12
δ (INT8)	≈ 2016
FP baseline PPL	≈ 10.5
Cal. samples	128–1024

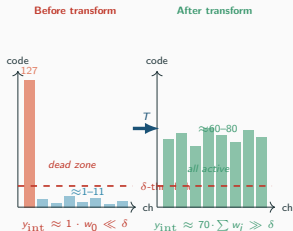
Why FlatQuant Transforms Matter for ADC

Without transform: sparse codes

- Per-token scale $s = \max(|x_{\text{tile}}|)/127$
- One large feature gets code ≈ 127 , rest get codes $\approx 1-10$
- Dot product: $y_{\text{int}} \approx 1 \cdot w_{\text{max}} \approx \text{small} \Rightarrow \text{dead zone}$

With FlatQuant transform

- Kronecker-decomposed orthogonal T learned per layer
- After rotation: all features have similar magnitude
- Codes $\approx 50-100$ for all channels
- $y_{\text{int}} \approx 64 \cdot \sum w_i \gg \delta \Rightarrow \text{no dead zone}$



Key proxy

$\text{PPL}_{\text{bypass}}$ = quality of the transforms.

Lower $\text{PPL}_{\text{bypass}} \Rightarrow \text{more uniform codes} \Rightarrow \text{less dead zone.}$

First Key Discovery: Dead Zone Was Not Hardware-Fundamental

Early baseline (branch llama-flatquant-adc)

- $PPL_{\text{bypass}} = 21.6$, $PPL_{\text{ADC}} = 28.99$
- $\text{dead_rate} = 81\%$
- PACT, penalties, L1 loss $\Rightarrow$ all failed

Improved baseline (branch pact)

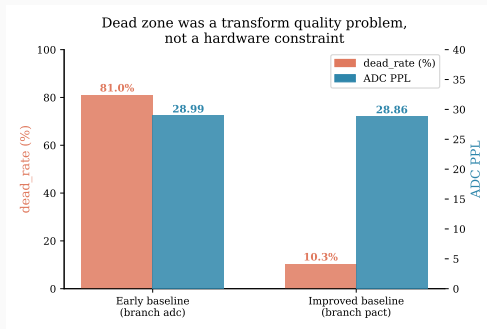
- $PPL_{\text{bypass}} = 10.0$, $PPL_{\text{ADC}} = 28.86$
- $\text{dead_rate} = 10.3\%$
- **Hardware unchanged.** Only better FlatQuant training (percentile calibration vs. absmax)

Implication:

Dead-zone collapse was a **transform quality problem**, not a fundamental ADC hardware constraint.

Remaining INT8 gap $PPL_{\text{bypass}}=10 \rightarrow PPL_{\text{ADC}}=29$:

ADC resolution — empirically only ~ 15 – 30 useful bins in $\pm\sigma_z$ (INT8, $\delta \approx 2016$).



INT8: Propagated Calibration Fixes the Real Problem

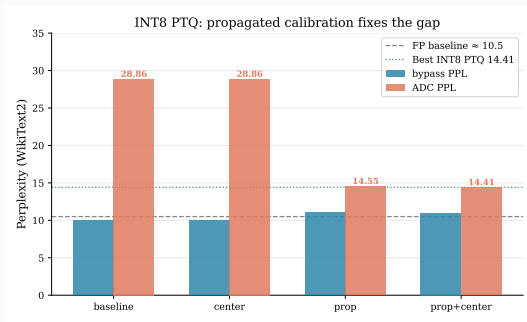
Two ideas for closing the 10 $\rightarrow$ 29 gap:

1. **Bin-centre loss** $\cos^2(\pi z)$:
nudge y_{int} toward bin centres.
Result: no improvement.
2. **Propagated calibration**:
train layer i on *ADC-quantised* inputs from layer $i - 1$
(not clean FP inputs).
Result: -47% ADC PPL.

Why propagation works

Standard layer-wise PTQ with FP inputs *underestimates* the reconstruction difficulty at inference, where upstream ADC errors accumulate across layers.

Method	PPL _{bypass}	PPL _{ADC}
baseline	10.01	28.86
center	10.01	28.86
prop	11.01	14.55
prop+center	11.00	14.41



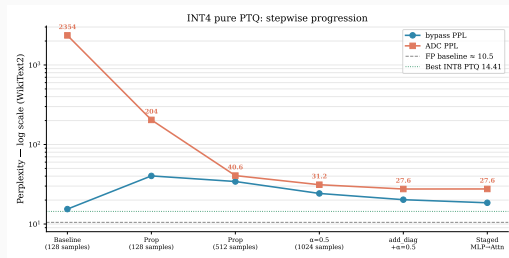
INT4 Pure PTQ: A Stepwise Story

Each step fixes one failure mode:

1. **Baseline INT4 (128 samples):** $\text{PPL}_{\text{ADC}} = 2355$ — catastrophic; 128 samples insufficient.
2. **Propagation + 1024 samples:** $\text{PPL}_{\text{ADC}} = 40.6$ — scale and propagation are the dominant levers.
3. **Partial propagation $\alpha = 0.5$:** $\text{PPL}_{\text{ADC}} = 31.2$ then **diag scaling** $\rightarrow 27.56$ — mixed FP+quant loss + bounded diagonal.
4. **Staged training (MLP $\rightarrow$ attn diag):** $\text{PPL}_{\text{ADC}} = 27.60$, $\text{PPL}_{\text{bypass}} = 18.50$ — cleanest bypass yet.

Key insight

Bin-centre loss is **harmful** for INT4: fine ADC step ($\delta \approx 6.12$) makes the penalty dominate and destabilise training.



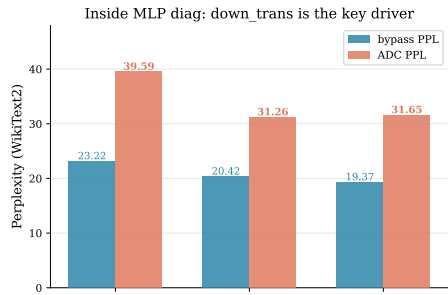
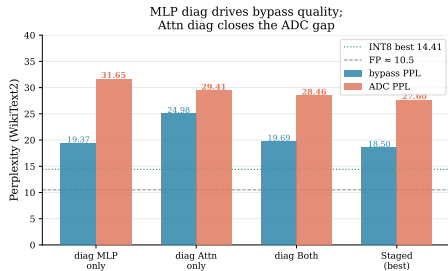
Mechanistic Insight: MLP and Attention Play Different Roles

Finding (v3/v4 ablation):

- **MLP diag** → best *bypass* PPL, but wide ADC gap (better transforms, less robust to ADC noise)
- **Attn diag** → does *not* improve bypass, but *closes* the ADC gap
- **Inside MLP diag**: `down_trans` is the key driver, not `up_gate_trans` (`down_proj` sits right before ADC accumulation)

Staged training exploits this:

1. Stage A (no prop): learn clean MLP diag transforms
2. Stage B (10 ep, $\alpha=0.5$): add attn diag, fine-tune for ADC robustness



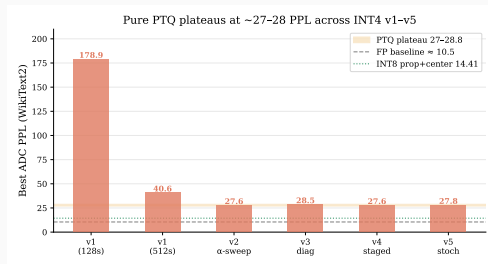
Pure PTQ Hits a Plateau — a Different Tool Is Needed

After all PTQ improvements (v2 through v5):

- Best $\text{PPL}_{\text{ADC}} \approx 27.5\text{--}28.5$ across all INT4 variants
- Adding more calibration samples / stochastic propagation / Hadamard gives < 0.5 PPL gain
- **Root cause:** 4-bit weight quantisation introduces systematic residual error that transform calibration cannot recover — transforms redistribute information, but cannot add it back after INT4 rounding

Key realisation

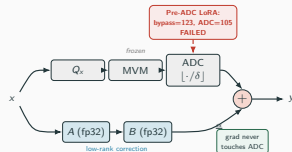
We need a **post-hoc correction** that can learn to model the ADC residual *after* quantisation, without re-training the frozen PTQ checkpoint.



Post-ADC Residual LoRA: Breaking the Plateau

Architecture:

$$y = \underbrace{\text{ADC}_{\text{frozen}}(x)}_{\text{PTQ path}} + \frac{\alpha}{r} B(A(x_{\mathbb{R}}))$$



- Correction applied **after** ADC output
- Base model frozen in fp16; LoRA adapters in fp32
- Gradients never touch round_ste / ADC clamp $\Rightarrow$ stable
- Trained with LM cross-entropy (+ optional KL teacher), 5 epochs

Pre-ADC LoRA failed

pre_adc_r4_down.o: bypass=123, ADC=105.

Gradients through `rnd_ste(W+ΔW)` are too noisy from step 0.

Best result (r4_all_ce_kl):

PTQ base	27.60
+ LoRA (all 7, r=4, CE+KL)	13.96

ADC-LoRA Ablations: Key Findings

1. CE+KL consistently beats CE

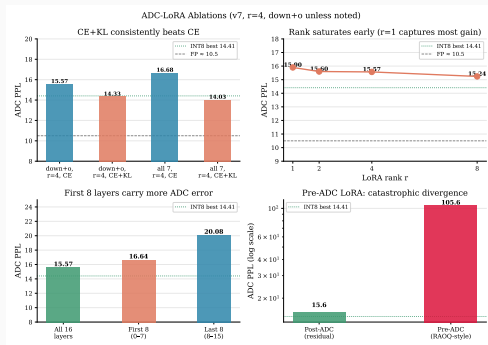
- down+o, r4: CE $\rightarrow$ 15.57; CE+KL $\rightarrow$ **14.33**
- all 7, r4: CE $\rightarrow$ 16.68; CE+KL $\rightarrow$ **14.03**
- FP teacher provides signal CE alone cannot

2. Early layers carry more ADC error

- first8: 16.64 last8: 20.08
- Early layers accumulate and *propagate* errors downstream
- All 16 layers (r=4, CE+KL): **13.96** — best result

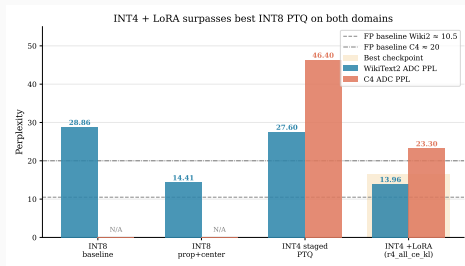
3. Post-ADC is the only stable placement

- Pre-ADC LoRA: bypass=123, ADC=105 — catastrophic failure
- Gradients through round_ste are too noisy from step 0
- Post-ADC: stable from epoch 1



Final Results: INT4 + LoRA Surpasses INT8 PTQ

Method	Bits	Wiki2	C4
FP baseline	—	~10.5	~20.0
INT8 baseline	w8a8	28.86	—
INT8 prop+center	w8a8	14.41	—
INT4 pure PTQ best	w4a4	27.56	—
INT4 staged PTQ	w4a4	27.60	46.40
INT4 + LoRA	w4a4	13.96	23.30

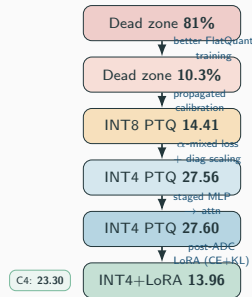


Key messages

- **INT4 + LoRA surpasses best INT8 PTQ on WikiText2** (13.96 vs. 14.41)
- **Generalises to C4** (−50% vs. INT4 PTQ baseline): correction is not overfit to calibration domain
- CE+KL with all 7 projections is the winning config

Takeaways

1. **Dead zone was a training problem, not hardware.**
Better FlatQuant calibration: 81% $\rightarrow$ 10.3% dead rate.
2. **Propagated calibration fixes INT8.**
ADC PPL: 28.86 $\rightarrow$ 14.41 (−47%), near FP baseline.
3. **Staged PTQ closes the INT4 gap to $\times 1.4$.**
 α -mixed loss + MLP-first diagonal training.
4. **MLP drives bypass; attn drives ADC robustness.**
Staged training combines both — insight, not just tuning.
5. **Post-ADC LoRA breaks the PTQ plateau.**
INT4 ADC 13.96 < INT8 PTQ 14.41. Generalises to C4 (−50% vs PTQ baseline).



*The right place to correct ADC error is **after** ADC, not before it.*

Backup: Complete INT8 Results (v2)

Experiment	Description	PPL _{bypass}	PPL _{ADC}	dead_rate
baseline	FlatQuant w8a8	10.01	28.86	10.3%
center mid	bin-centre $\lambda = 0.1$	10.01	28.86	10.3%
prop	propagated calibration	11.01	14.55	10.3%
prop+center	propagated + bin-centre	11.00	14.41	10.3%

Backup: Complete INT4 v2 Alpha-Sweep

Experiment	α	PPL _{bypass}	PPL _{ADC}	gap
baseline_1024	0.0	19.80	56.83	×2.87
prop_full_1024	1.0	30.32	40.20	×1.32
propalpha_075_1024	0.75	27.46	35.92	×1.31
propalpha_05_1024	0.5	24.24	31.22	×1.29
propalpha_025_1024	0.25	21.65	32.35	×1.49
add_diag + $\alpha=0.5$	0.5	20.23	27.56	×1.36
staged_mlp_attn	0.5	18.50	27.60	×1.49

Backup: Complete v7 LoRA Ablation

Experiment	Targets	Loss	Rank	PPL _{bypass}	PPL _{ADC}
base (no LoRA)	—	—	—	17.89	27.55
r1_down_o	down+o	CE	1	—	15.90
r2_down_o	down+o	CE	2	—	15.60
r4_down_o	down+o	CE	4	18.68	15.57
r8_down_o	down+o	CE	8	—	15.24
r4_down_o_ce_kl	down+o	CE+KL	4	19.54	14.33
r4_down_o_first8	down+o	CE	4	18.42	16.64
r4_down_o_last8	down+o	CE	4	18.62	20.08
pre_adc_r4_down_o	down+o	CE	4	123.34	105.63
r4_all_ce	all 7	CE	4	17.62	16.68
r4_all_ce_kl	all 7	CE+KL	4	19.13	14.03

Backup: v8 Generalisation to C4

Experiment	Targets	Wiki bypass	Wiki ADC	C4 bypass	C4 ADC
base (no LoRA)	—	18.49	27.26	29.41	46.40
rank8_down_o (CE)	down+o	18.51	15.47	30.48	29.39
r4_down_o_ce_kl	down+o	19.05	14.27	29.37	23.88
r4_all_ce_kl	all 7	18.70	13.96	29.10	23.30

Backup: Stochastic Propagation (v5)

Experiment	Mode	PPL _{bypass}	PPL _{ADC}	gap
stoch_bern (flat)	Bernoulli	19.25	35.38	×1.84
stoch_beta2 (flat)	Beta(2,2)	22.90	31.40	×1.37
staged + Bernoulli	Bernoulli	16.94	32.44	×1.92
staged + Beta(2,2)	Beta(2,2)	17.16	27.82	×1.62
deterministic $\alpha=0.5$	—	24.24	31.22	×1.29

Bernoulli gives best bypass but worst ADC gap (hard 0/1 switching provides no ADC signal on FP batches). Beta(2,2) $\approx$ deterministic $\alpha=0.5$ in expectation.

Backup: Hardware Config & Delta Formula

Parameter	Value
b_w (weight bits)	4 / 8
b_x (act. bits)	4 / 8
b_a (ADC bits)	8
<code>mvm_limit</code>	256 / 1024
k (ADC parallelism)	16
δ (mvm=256, INT8)	≈ 2016
δ (mvm=256, INT4)	≈ 6.12
δ (mvm=1024, INT8)	≈ 8064
FlatQuant epochs	30
FlatQuant LR	0.005
Cal. samples (INT4)	1024
Cal. samples (INT8)	128

Delta formula ($q = 2^{b-1} - 1$):

$$\delta = \frac{2 \cdot \text{tile_in} \cdot q_x \cdot q_w}{2^{b_a} \cdot k}$$

- INT8 ($q=127$, mvm=256): $\frac{2 \cdot 256 \cdot 127^2}{256 \cdot 16} \approx 2016$
- INT4 ($q=7$, mvm=256): $\frac{2 \cdot 256 \cdot 7^2}{256 \cdot 16} \approx 6.12$
- INT8, mvm=1024: $\frac{2 \cdot 1024 \cdot 127^2}{256 \cdot 16} \approx 8064$

INT8: useful ADC bins in $\pm \sigma_z \approx 15\text{--}30$ (empirical) — explains why INT8 PTQ alone plateaus.

Loss Landscape: Why $\alpha=0.5$ Is the Sweet Spot

Dual-forward mixing loss:

$$\mathcal{L} = (1-\alpha)\mathcal{L}_{\text{FP}} + \alpha\mathcal{L}_{\text{ADC}}$$

- $\alpha=0$: clean FP loss only $\Rightarrow$ flat bypass, wide ADC gap
- $\alpha=1$: full ADC loss $\Rightarrow$ transforms over-specialise for noise, bypass PPL degrades sharply
- $\alpha=0.5$: smooth, wide minimum in both $\mathcal{L}_{\text{bypass}}$ and $\mathcal{L}_{\text{ADC}}$

Visualisation (Li et al. "Filter Normalisation" method):

1. Train FlatQuant to convergence at $\alpha \in \{0, 0.5, 1\}$
2. Perturb $\theta^* + a\mathbf{d}_1 + b\mathbf{d}_2$ (filter-normalised random directions)
3. Plot $\text{PPL}_{\text{ADC}}(a, b)$ as a 2D surface / heatmap
4. Compare sharpness of minima across α

Expected result

$\alpha=0.5$ should show the widest, flattest minimum — both FP and ADC loss curvature are moderate.

TODO figure

3 panels (one per α): 2D contour heatmap of ADC PPL as a function of perturbation (a, b) around θ^* . Color: low PPL = blue (flat/wide), high PPL = red (sharp). Panel 1 ($\alpha=0$): narrow deep valley in bypass dim, flat in ADC dim. Panel 2 ($\alpha=0.5$): wide flat basin in both dims. Panel 3 ($\alpha=1$): very sharp minimum (overfit to ADC noise). See `experiments/loss_landscape.md` for run instructions.

This Work in Context

Method	Key idea	Gap / limit
FlatQuant	Learned Kronecker-decomposed orthogonal transforms per layer	Calibrated on FP loss only; no ADC signal
OmniQuant / LET	Learnable per-channel equivalent scaling (diagonal)	No ADC-aware objective
QEP / propagated calibration	Feed quantised activations into the next layer's calibration	INT8 only; no mixed-loss
PACT / QDrop	Stochastic / clamped quant-aware training	Full re-training required; too expensive here
This work	ADC-aware PTQ (α -mixed loss, staged diag) + post-ADC LoRA correction	—

Key novelty: closing the loop between the *hardware ADC model* and the *calibration / fine-tuning objective* at every stage.

No prior method directly targets the **post-ADC residual**; our LoRA correction is the first to do so.

Post-ADC LoRA: Parameter and Compute Overhead

Trainable parameter count ($r=4$, all 7 projections):

Projection	A	B
q_proj	4×2048	2048×4
k_proj	4×2048	256×4
v_proj	4×2048	256×4
o_proj	4×2048	2048×4
gate_proj	4×2048	8192×4
up_proj	4×2048	8192×4
down_proj	4×8192	2048×4
Per layer		174 K
$\times 16$ layers		2.8M
Base model (1B)		0.28%

Inference overhead (per forward pass):

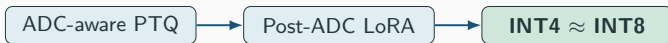
- Extra FLOPs per projection: $d_{\text{in}} \cdot r + r \cdot d_{\text{out}} \ll d_{\text{in}} \cdot d_{\text{out}}$
- Example: q_proj: $8192+8192$ vs. $4M$ ($< 0.5\%$ overhead)
- Memory: 2.8M fp32 params $\approx$ **11 MB** vs. ~ 2 GB base model
- LoRA correction runs on *digital* hardware — does not increase analog area

Bottom line

0.28% extra params, $< 0.5\%$ extra FLOPs, correction entirely in digital post-processing. The trade-off is strongly favourable.

Thank you!

Questions?



The right place to correct ADC error is after ADC, not before it.